The Transformer Architecture: Attention Is All You Need

In 2017, Vaswani et al. published the seminal paper 'Attention Is All You Need', introducing the Transformer architecture. Before the Transformer, sequence transduction models relied heavily on complex recurrent neural networks (RNNs) or convolutional neural networks (CNNs) in encoder-decoder structures. These models faced significant limits: recurrent networks process sequences sequentially, which prevents parallelization during training and makes it difficult to model long-range dependencies.

1. The Core Innovation: Self-Attention Mechanism

The fundamental innovation of the Transformer is the self-attention mechanism. Unlike recurrent networks that pass a hidden state step-by-step, self-attention allows the model to relate different positions of a single sequence directly. For each word (token) in the input, the model computes three vectors: Query (Q), Key (K), and Value (V). The attention weights are calculated using the Scaled Dot-Product Attention formula: $\mathrm{Attention}(Q, K, V) = \mathrm{softmax}((Q * K^T) / \mathrm{sqrt}(d_k)) * V$ where d_k is the dimension of the keys. This scaling factor prevents the dot products from growing excessively large, which would push the softmax function into regions with extremely small gradients.

2. Multi-Head Attention

Instead of performing a single attention function, the Transformer utilizes Multi-Head Attention. This involves linearly projecting the Queries, Keys, and Values h times with different, learned projections. The attention function is run in parallel on each projection, and the resulting outputs are concatenated and projected again. This allows the model to jointly attend to information from different representation subspaces at different positions.

3. Encoder-Decoder Structure

The encoder consists of a stack of N=6 identical layers. Each layer has two sub-layers: a multi-head self-attention mechanism and a simple, position-wise fully connected feed-forward network. Residual connections are applied around each sub-layer, followed by layer normalization. The decoder also contains a stack of N=6 identical layers, but inserts a third sub-layer which performs multi-head attention over the output of the encoder stack. The self-attention sub-layers in the decoder are masked to prevent positions from attending to subsequent positions (preserving auto-regressive properties).

4. Positional Encodings

Since the Transformer contains no recurrence or convolution, it has no inherent sense of the order or position of tokens in the sequence. To inject positional information, the authors add 'positional encodings' to the input embeddings. These encodings utilize sine and cosine functions of different frequencies, allowing the model to easily learn to attend by relative positions.